CHRONOS

TESTING AND RESULTS

Alarm Clock and Time Management Dashboard

A record of the Ubuntu build and automated HTTP integration smoke test

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1. PURPOSE AND SCOPE

This document records the verification run performed for Chronos, a local browser dashboard backed by a C/POSIX service on Ubuntu. It supplements the full project report with the actual build/test command, observed output, test-case results and the limits of what the automated test establishes.

The tested smoke test starts a real local HTTP service on an available loopback port, uses an isolated temporary directory for runtime data, sends HTTP requests, waits for scheduler events, restarts the server and checks persisted state. The results below are from one fresh forced build and smoke-test run on Ubuntu under WSL.

2. TEST ENVIRONMENT

Item	Observed configuration
Operating system	Ubuntu Linux under Windows Subsystem for Linux (WSL)
Compiler	GCC; C11, optimization and warning flags
Build tool	GNU Make
Test runner	Python 3 standard library
Service endpoint	127.0.0.1 on a dynamically selected free port
Alarm/history data	Isolated temporary directory; no project data reused
Execution command	make -B test

3. TEST METHOD

The command make -B test forces GNU Make to rebuild the C executable even if a previous binary exists, then runs tests/smoke_test.py against that binary. The test script is assertion-based: it stops and returns a non-zero exit if a checked condition fails. A successful end-of-test message is printed only after the assertions complete.

```
cc  -std=c11 -O2 -Wall -Wextra -Wpedantic src/main.c  -pthread -o build/chronos
python3 tests/smoke_test.py ./build/chronos
Smoke test passed: UI, map, alarm firing/snooze, restart persistence, and API.
```

4. TEST CASES AND RESULTS

The table groups assertions from the single integration test into functional test cases. PASS means all assertions in that group passed in this execution; this is not a claim of repeated or long-duration testing.

ID	Test case	Expected / verified result	Result
T01	Forced Ubuntu build	C source compiles with configured warning flags.	**PASS**
T02	HTTP service startup	Service becomes ready on an available loopback port.	**PASS**
T03	Dashboard assets	HTML, CSS and JavaScript routes return expected UI features.	**PASS**
T04	UI requirements	Map, stop/snooze controls, weekday choices and history appear; removed system panel/footer are absent.	**PASS**
T05	Initial API state	State endpoint returns valid JSON with no alarms and no system-monitor payload.	**PASS**
T06	Alarm lifecycle	Create, disable with toggle, and delete return expected results.	**PASS**
T07	Invalid pending snooze	Snoozing an alarm that has not fired is rejected with HTTP 409.	**PASS**
T08	Weekly alarm setup	Repeat-day mask and time zone are accepted and returned by the API.	**PASS**
T09	Weekly firing and recurrence	Alarm fires; remains enabled and retains its repeat mask with a future weekly time.	**PASS**
T10	Weekly snooze	Snooze is accepted and does not erase the recurring schedule.	**PASS**
T11	One-time firing	Forked scheduler fires the alarm, disables it and records a history event.	**PASS**
T12	One-time snooze	Fired alarm is re-enabled with its due time moved at least 295 seconds forward.	**PASS**
T13	Restart persistence	Alarm, recurrence/snooze fields and firing history are restored after service restart.	**PASS**
T14	History deletion	Deleting the occurrence succeeds, removes it, and a repeated delete returns HTTP 404.	**PASS**

5. RESULT ANALYSIS

The forced compilation completed successfully with the project's configured C11 and warning options. The Python smoke test then completed and printed its success line. Taken together, this is evidence that the tested source revision builds and that the listed local HTTP, scheduler, alarm, snooze, persistence and history flows worked in the test environment.

Measure	Observed result
Build	PASS — make -B test rebuilt build/chronos.
Automated test suite	PASS — smoke_test.py exited successfully.
Grouped test cases	14 of 14 passed in the completed smoke-test run.
Process cleanup	Smoke test uses finally cleanup for its test service.

6. BEHAVIOURAL DETAILS VERIFIED

The service-level tests verify that firing an alarm is not merely a front-end display change: the scheduler event reaches the parent, appears in API history and updates persisted alarm state. The weekly case confirms a recurrence remains enabled after firing and keeps a future scheduled timestamp. Snoozing a recurring alarm preserves its weekday mask; the one-time snooze moves the due time forward. The restart case checks that these values and the firing history can be loaded by a new parent process.

History deletion is separately checked from alarm deletion: it removes only the selected occurrence from history. A second attempt against the already-deleted entry must report 404. A pending alarm cannot be snoozed and is expected to return 409.

7. LIMITATIONS AND INTERPRETATION

This is an integration smoke test, not exhaustive verification. Its coverage and boundaries should be stated accurately:

Area	What this test demonstrates	Not demonstrated
Alarm audio	Checks that browser source includes the repeating audio and stop/snooze handlers.	Does not measure sound or prove audible output on a physical device.
Browser UI	Checks served markup/script requirements and expected element/function presence.	Does not automate real browser interaction, accessibility or all viewport layouts.
Scheduler	Checks near-future one-time and weekly alarms, firing state and snooze responses.	Does not test long-term reliability, every time zone or daylight-saving transition.
Runtime	Checks service restart and persisted state while the computer is active.	Does not test system suspend, power loss or abrupt storage failure.
Security/scale	Uses a local test endpoint and isolated test data.	Does not assess public-network exposure, multi-user access, load or production hardening.

8. CONCLUSION

The fresh Ubuntu/WSL build and automated Chronos smoke test both passed. All 14 grouped checks recorded in this report passed during the run, including one-time and recurring alarm firing, snooze, service restart persistence and alarm-history deletion. These results validate the tested local prototype behaviours; the limitations above remain outside the scope of the automated smoke test.

APPENDIX A — REPRODUCING THE TEST

Prerequisites

Open the project in an Ubuntu/WSL terminal and ensure GCC, GNU Make and Python 3 are installed. Change to the project root before running the command.

```
cd /mnt/c/Users/HP/OneDrive/Desktop/Os
make -B test
```

Expected successful output

```
cc  -std=c11 -O2 -Wall -Wextra -Wpedantic src/main.c  -pthread -o build/chronos
python3 tests/smoke_test.py ./build/chronos
Smoke test passed: UI, map, alarm firing/snooze, restart persistence, and API.
```

Run these commands from an Ubuntu/WSL terminal opened at the project directory. The test chooses a free local port and temporary alarm data directory automatically. A successful run ends with the smoke-test passed message shown in Section 3.